

Coffee Shop Sales Analysis

Introduction:

At the beginning of this BAN 510 project, I opened the Excel file to get a quick understanding of the dataset. It included coffee shop sales data with details about orders, customers, products, locations, and payment methods. With several hundred rows and a good mix of transactional and customer-level information, the dataset offered enough variety to explore overall sales performance and business trends.

To start, I reviewed the number of observations, looked through the columns, and applied filters to identify unusual or inconsistent entries. For example, filters were used on the city column to find spelling differences or extra spaces. This approach helped make the data cleaning process faster and more accurate.

Power Query was used to clean the data by trimming extra spaces, correcting column formats, and removing duplicates to prepare it for analysis. After cleaning, I explored possibilities for transforming the data further. There was scope to create many new variables, but I only transformed the ones that made sense and could lead to meaningful visuals and insights. Examples include drink size, profit margin category, and customer age group, all of which supported clearer analysis and business understanding.

The cleaned dataset was then split into three separate tables: Orders, Customers, and Products. Primary keys were added to each table, and relationships were created to connect them properly. After building the data model, the entire structure was loaded into Power BI. From there, visualizations were developed to explore patterns, trends, and key business insights.

Coffee Shop Sales Analysis By Charts

1. Pie Chart (Profit and Total Sales by Product Type)

The analysis began with a broad overview of sales and profit across product categories. This chart compared affordable, basic, and premium drinks. Basic drinks generated the highest profit of 1.32K, which made up 53.76 percent of total profit and 1,729 total sales. Premium drinks followed with a profit of 1,016 and 1,653 in sales. Affordable drinks contributed only 120 in profit and had just 22 sales.

This helped identify where the business is actually earning profit. Even though affordable drinks were priced low, they made a very small impact. The focus should remain on basic and premium products since they contribute almost all of the profit and volume. This set the base for understanding where the real value lies.

2. Clustered Column Chart (Sales Volume by Coffee Type and Order Size)

After identifying high-performing product types, I explored how order size and customer preferences impacted sales. This chart looked at coffee types such as drip coffee, cold brew, mocha, flat white, and cappuccino, each grouped by order size. Drip coffee in large size had the highest number of orders with 30. Cold brew was mostly ordered in medium size, and cappuccino was most popular in small size.

An age group filter was applied to explore how size preferences change across customer segments. These details gave more depth to the product insights, showing not just what sells, but how different age groups interact with order size and product type. This can guide promotions for specific groups or help plan upselling strategies.

3. Treemap (Sales Breakdown by Coffee Type and Drink Size)

This chart continued the analysis by focusing on the combination of drink type and drink size, with a filter applied for profit margin. Drip coffee in medium size had the highest number of sales. Mocha was most commonly ordered in large size, while flat white, cold brew, and cappuccino were mainly sold in small sizes.

The profit margin filter revealed that many high-selling combinations belonged to low or medium profit tiers. This helped highlight that popularity does not always mean profitability. The business can use this insight to adjust pricing or focus more on promoting combinations that fall into the high-margin category.

4. Area Chart (Daily Trends in Sales Cost, Profit, and Order Volume)

This chart added a time-based view, showing changes in profit, cost, and order volume over individual days. For example, the lowest sales occurred on January 16, with just 3 orders, a cost of 4.75, and a profit of 17.75. On January 12, there were 19 orders, a cost of 73, and a profit of 169.50.

These patterns can help the business understand slow days and peak performance periods. It also connects with earlier visuals by showing how the timing of high-margin items and customer behavior affects daily performance. This kind of trend analysis supports inventory planning, promotion timing, and seasonal forecasting.

5. Line and Clustered Column Chart (Orders and Average Profit by Payment Method)

This chart examined how customers pay and how each payment method affects profitability. Gift cards had the highest average profit per order at 8.57. Cash followed with 68 orders and an average profit of 7.88. Google Pay had 49 orders and the lowest average profit of 7.14.

A customer age group filter was used to better understand payment habits across different groups. This connected to earlier customer behavior insights and showed how certain methods bring in more value. Promoting gift card usage could increase average profit, while

understanding the lower return from some digital payments can guide future pricing or marketing strategies.

6. Stacked Bar Chart (Sales by City and Age Group)

The final chart introduced a location-based view. It showed total sales by city, stacked by age group. In Flagstaff, all sales came from young professionals, totaling 760. In Reno, the lowest number of sales came entirely from working adults, with just 56. A quick glance at the chart showed that young professionals made the majority of purchases across all cities.

This confirmed the importance of targeting young professionals. They are the most active customer group and should be the focus when planning promotions, building loyalty programs, or choosing locations for expansion.

Conclusion:

The analysis revealed that not all high-selling products bring high profits, highlighting the importance of focusing on product combinations that offer both volume and value. Size preferences varied by drink type and customer age, which can guide targeted upselling and bundling strategies. Payment method analysis showed clear differences in profitability, making it worth exploring incentives for high-return options like gift cards.

Time-based trends pointed to daily fluctuations that can be used to schedule promotions and optimize staffing. City-wise sales, when paired with customer demographics, showed that specific regions are driven by very distinct customer groups. Young professionals emerged as the most consistent and valuable segment across cities.

By aligning product strategy with customer behavior, payment trends, and regional patterns, the business can take more focused and confident decisions. Emphasizing high-margin drink sizes, refining payment method strategies, and promoting in cities with engaged customer bases will help maximize returns.

Final Line: Targeting young professionals with high-margin drink combinations and promoting gift card payments offers the greatest potential for boosting both sales and profitability.